

Introduction to Bonds

A bond is a type of debt security, similar to a loan, where an investor lends money to an entity (typically a corporation or government) which borrows the funds for a defined period of time at a variable or fixed interest rate. Bonds are used by companies and governments to finance projects and operations.[8]

Key Components of a Bond

- **Face Value (or Par Value):** This is the amount of money the bond will be worth at its maturity. It is also the reference amount the bond issuer uses when calculating interest payments. Most often, the face value is \$1,000.
- **Coupon Rate:** The coupon rate is the rate of interest the bond issuer pays to the bondholder. This rate is a percentage of the face value and is paid out at specified intervals (e.g., semiannually or annually).
- **Coupon Payments:** These are the periodic interest payments made to the bondholder. The amount is calculated by multiplying the face value by the coupon rate.
- **Maturity Date:** This is the date on which the bond's final payment is due, and the principal (face value) is repaid to the bondholder. The bond is said to have "matured" at this point.
- **Issuer:** The entity that borrows the money and is responsible for repaying the principal and making interest payments.

Types of Bonds

There are many types of bonds, but they can be broadly categorized based on their payment structure:

- **Coupon Bonds:** These bonds make regular interest payments (coupons) to the bondholder throughout the life of the bond, in addition to repaying the face value at maturity. The provided text focuses on pricing this type of bond. [5]
- **Zero-Coupon Bonds:** These bonds do not pay periodic interest. Instead, they are issued at a significant discount to their face value and repay the full face value at maturity. The investor's return is the difference between the purchase price and the face value. "Strips" are a type of synthetic zero-coupon bond mentioned in the text.[4][6]

4.4 Bond Pricing

The theoretical price of a bond is the present value of all its future cash flows (coupons and principal). A more accurate pricing method uses a different zero rate to discount each individual cash flow according to its maturity.

Example:

Consider a 2-year Treasury bond with a \$100 principal and a 6% annual coupon paid semiannually. The cash flows are \$3 every six months for two years, plus a final principal payment of \$100. Using the zero rates from Table 4.2 in the text:

Maturity (Years)	Zero Rate (Continuous)
0.5	5.0%
1.0	5.8%
1.5	6.4%
2.0	6.8%

The theoretical price is calculated as:

$$\text{Price} = 3e^{-0.05 \times 0.5} + 3e^{-0.058 \times 1.0} + 3e^{-0.064 \times 1.5} + 103e^{-0.068 \times 2.0} = \$98.39$$

Bond Yield

A bond's yield (or Yield to Maturity, YTM) is the single discount rate (y) that, when applied to all its cash flows, equals the bond's market price.

For the bond above with a market price of \$98.39, the yield y is the solution to:

$$3e^{-y \times 0.5} + 3e^{-y \times 1.0} + 3e^{-y \times 1.5} + 103e^{-y \times 2.0} = 98.39$$

This equation is solved iteratively and gives a yield $y = 6.76\%$ with continuous compounding.

Par Yield

The par yield is the coupon rate that makes a bond's price equal to its par value (e.g., \$100). The formula to find the par yield c is:

$$c = \frac{(100 - 100d)m}{A}$$

Where:

- m = number of coupon payments per year.
- d = present value of \$1 received at maturity.
- A = present value of an annuity of \$1 paid on each coupon date.

Using the zero rates from the example, the 2-year par yield with semiannual coupons is 6.87%.

4.5 Determining Treasury Zero Rates (Bootstrapping)

The bootstrap method is used to determine Treasury zero rates from the prices of coupon-bearing bonds. It is an iterative process that starts with short-term zero-coupon bonds and sequentially calculates longer-term zero rates.

Example of the Bootstrap Method

Using the data from Table 4.3 in the text:

Principal	Maturity (Yrs)	Annual Coupon	Price
\$100	0.25	0	\$97.5
\$100	0.50	0	\$94.9
\$100	1.00	0	\$90.0
\$100	1.50	\$8	\$96.0
\$100	2.00	\$12	\$101.6

Step 1: Calculate 3-Month (0.25-year) Zero Rate

$$100 = 97.5e^{R \times 0.25} \implies R = 10.127\%$$

Step 2: Calculate 6-Month (0.50-year) Zero Rate

$$100 = 94.9e^{R \times 0.5} \implies R = 10.469\%$$

Step 3: Calculate 1-Year Zero Rate

$$100 = 90.0e^{R \times 1.0} \implies R = 10.536\%$$

Step 4: Calculate 1.5-Year Zero Rate

The 1.5-year bond pays \$4 at 6 months and 1 year, and \$104 at 1.5 years. We use the rates from prior steps to discount the known cash flows:

$$96 = 4e^{-0.10469 \times 0.5} + 4e^{-0.10536 \times 1.0} + 104e^{-R \times 1.5}$$

Solving for R , the 1.5-year rate, gives $R = 10.681\%$.

Step 5: Calculate 2-Year Zero Rate

The 2-year bond pays \$6 at 6 months, 1 year, and 1.5 years, and \$106 at 2 years. Using the previously found rates:

$$101.6 = 6e^{-0.10469 \times 0.5} + 6e^{-0.10536 \times 1.0} + 6e^{-0.10681 \times 1.5} + 106e^{-R \times 2.0}$$

Solving for R , the 2-year rate, gives $R = 10.808\%$.

This process constructs the **zero curve**, which plots the zero rate as a function of maturity.